

AI Exam Tutorials — Full Answer Key

Introduction to Artificial Intelligence · Level 3 · model answers to every question in AI_EXAM_TUTORIALS.pdf

Answers are written to be exam-ready (concise but complete). Use alongside the main "Introduction to AI — Exam Revision Pack" for the underlying theory.

Module 1 — Foundations, History & Philosophy (Q1-Q7)

Q1. Define AI; the four approaches; which is most adopted.

AI is the branch of computer science that builds systems able to perform tasks normally requiring human intelligence. The four approaches (Russell & Norvig): **Acting humanly** (Turing-Test approach — behave like a human); **Thinking humanly** (cognitive modelling — replicate human thought); **Thinking rationally** (laws of thought — correct logical reasoning); **Acting rationally** (rational agent — act to best achieve goals). **Most adopted: acting rationally (rational agent)** — it is general, mathematically well-defined, measurable, and handles uncertainty rather than depending on imitating humans.

Q2. History from 1956 to today; two AI winters; the AI spring.

1956 Dartmouth Conference coins "AI"; optimism era (search, early NLP, perceptrons). **First AI winter (~1974-80)**: Lighthill report, Minsky & Papert's perceptron limits, unmet promises → funding cuts. **1980s**: expert-systems boom. **Second AI winter (~1987-93)**: collapse of the LISP-machine market, expert systems too brittle/expensive. **1990s-2000s**: statistical ML and data-driven methods. **AI spring (2010s+)**: driven by **big data, GPU compute, and deep learning** (2012 AlexNet), now large language models.

Q3. Turing Test; significance; two criticisms.

Turing (1950) "imitation game": a judge holds text conversations with a human and a machine; if the judge cannot reliably tell them apart, the machine is deemed intelligent. **Significance**: first operational, behavioural definition of machine intelligence; shifted the question from "can machines think?" to "can they behave intelligently?". **Criticisms**: (1) it tests imitation/behaviour, not genuine understanding (Searle's Chinese Room); (2) it is anthropocentric and gameable — programs like ELIZA fool judges with tricks while ignoring non-verbal intelligence.

Q4. ANI vs AGI vs ASI; examples of ANI.

ANI (Narrow): excels at one specific task — **all AI today**. Examples: spam filters, Siri/Alexa, AlphaGo, facial recognition, Netflix recommendations. **AGI (General)**: human-level ability across any task (hypothetical). **ASI (Super)**: surpasses human intelligence in all domains (hypothetical/future).

Q5. Symbolic AI (GOFAI) vs Connectionist (neural).

GOFAI: explicit symbols, rules and logic. Strengths: interpretable, strong reasoning/planning, needs no data. Weaknesses: brittle, poor with noise/perception, knowledge-acquisition bottleneck. **Connectionist**: neural networks learning from data via weighted connections. Strengths: excellent for perception, noise and high-dimensional data, learns patterns automatically. Weaknesses: black-box, data-hungry, weak at explicit logical reasoning.

Q6. The Chinese Room argument (Searle).

A person who knows no Chinese sits in a room and produces correct Chinese replies purely by following an English rulebook to manipulate symbols. They pass for a Chinese speaker yet **understand nothing**. Searle argues that running a program is mere **syntax (symbol manipulation)**, which can never produce **semantics (genuine understanding, consciousness, intentionality)**. Hence passing the Turing Test does not prove real understanding — a challenge to "Strong AI".

Q7. Turing vs McCarthy.

Alan Turing — theory of computation, computability, machine learning and the behavioural Turing Test → the **computation/learning** trajectory (leading toward ML). **John McCarthy** — coined "AI" (1956), invented LISP, championed formal logic, knowledge representation and common-sense reasoning → the **symbolic/logic** trajectory (GOFAI/KR). Together they seeded the two long-running branches of AI research.

Module 2 — Intelligent Agents & Environments (Q8-Q12)

Q8. Intelligent agent; components of a rational agent; rationality.

An agent perceives its environment through **sensors** and acts on it through **actuators**. A rational agent is defined by its percept sequence, its knowledge of the environment, its possible actions, and a **performance measure**. **Rationality** = selecting the action expected to **maximize the performance measure** given the percepts and knowledge available — not omniscience, but doing the best possible given the information.

Q9. PEAS for an autonomous robotic surgical system.

P — Performance	E — Environment	A — Actuators	S — Sensors
Patient safety, precision, minimal tissue damage, success rate, speed	Operating room & patient's body (partially observable, dynamic, continuous)	Robotic arms, scalpels, sutures, cauterizer	Cameras, force/torque sensors, vital-sign monitors, imaging (ultrasound/MRI)

Q10. Fully vs partially observable + examples.

Fully observable: sensors reveal the complete state (chess). **Partially observable**: state is hidden or noisy, so the agent must keep an internal **belief state/memory** — e.g. a self-driving car (can't see around corners) or poker (opponents' cards hidden).

Q11. Deterministic vs stochastic.

Deterministic: the next state is fully fixed by the current state and action (chess). **Stochastic**: outcomes are probabilistic (dice, real-world robotics with sensor/actuator noise). Higher stochasticity pushes the agent from plain search toward **probabilistic reasoning / MDPs**.

Q12. Single vs multi-agent; cooperative vs competitive.

Single-agent: only one agent affects the outcome (solving a crossword). **Multi-agent:** other agents matter. **Cooperative** — shared goal (a fleet of self-driving cars avoiding collisions; warehouse robots). **Competitive** — opposing goals, often zero-sum (chess, poker).

Module 3 — Search (Q13-Q20)

Q13. Components of a well-defined search problem.

State space, **initial state**, **actions**, **transition model** (result of an action), **goal test**, and **path cost** (sum of step costs). A solution is an action sequence from initial state to a goal; the optimal solution has the lowest path cost.

Q14. BFS; when optimal; limits.

BFS expands the shallowest unexpanded node using a FIFO queue, exploring level by level. It is **complete**, and **optimal when all step costs are equal**. Its main weakness is memory: both time and space are $O(b^d)$ because the entire frontier must be stored.

Q15. DFS & iterative deepening; when to prefer DFS.

DFS expands the deepest node (stack); space is only $O(b \cdot m)$ — memory-efficient — but it is **not complete** in infinite spaces and **not optimal**. **Iterative Deepening DFS** runs depth-limited DFS with increasing limits, combining DFS's low memory with BFS's completeness/optimalty. Prefer DFS when solutions are deep, memory is tight, or the tree is finite.

Q16. Heuristic search; heuristic function; admissible vs consistent.

Informed search uses problem knowledge via a **heuristic $h(n)$** = estimated cost from n to the goal. **Admissible:** $h(n) \leq$ true cost (never overestimates) → guarantees A* optimality. **Consistent (monotonic):** $h(n) \leq c(n, n') + h(n')$ for every successor (triangle inequality); consistency implies admissibility.

Q17. A* step-by-step; optimal efficiency; $f = g + h$.

A* keeps a priority queue ordered by **$f(n) = g(n) + h(n)$** (g = cost so far, h = estimated cost to goal). It repeatedly expands the lowest- f node, generates successors, updates g , and tests the goal when a node is selected. It is **optimal** with an admissible (tree) / consistent (graph) heuristic, and **optimally efficient** — no other optimal algorithm expands fewer nodes for the same heuristic. f balances past cost with the estimate of future cost.

Q18. Hill-climbing vs simulated annealing.

Hill-climbing: greedy local search that always moves to a better neighbour; its drawback is getting stuck in **local maxima, plateaus and ridges**. **Simulated annealing:** sometimes accepts worse moves with probability $e^{-\Delta E/T}$; the **temperature T** starts high (more random moves) and cools over time, letting it escape local optima and converge toward the global optimum as $T \rightarrow 0$.

Q19. Adversarial search; Minimax.

Adversarial search handles competitive games with opposing agents. **Minimax:** MAX tries to maximize the score, MIN to minimize it; the algorithm recursively computes each node's value assuming both play optimally, and MAX picks the move with the highest guaranteed (minimum) value. It yields **optimal play** in zero-sum, deterministic, perfect-information games.

Q20. Alpha-Beta pruning.

Alpha-Beta keeps bounds α (best already guaranteed for MAX) and β (best for MIN); when $\alpha \geq \beta$ the remaining branch is pruned. It does **not change the result** because pruned branches could not influence the final minimax value. With good move ordering it cuts complexity from $O(b^d)$ to $O(b^{d/2})$, effectively doubling search depth.

Module 4 — Knowledge Representation & Reasoning (Q21-Q25)

Q21. KR; why critical; trade-offs of three formalisms.

KR is how knowledge about the world is encoded so a machine can reason with it — critical because intelligent behaviour needs structured, usable knowledge. **Logic:** precise and supports sound inference, but rigid and computationally costly. **Semantic nets:** intuitive graphs with inheritance, but informal/ambiguous semantics. **Frames:** structured objects with slots and defaults (good for stereotypes), but limited reasoning power.

Q22. Propositional vs First-Order Logic.

Propositional logic represents facts as true/false propositions — it cannot express objects, relations, or "all/some". **FOL** adds **objects, predicates/relations, functions and quantifiers ($\forall, \exists$)**, so "All humans are mortal" becomes one compact statement ($\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$). FOL is far more expressive and concise for complex domains.

Q23. Inference engine; forward vs backward chaining; when backward wins.

An **inference engine** applies rules to known facts to derive new facts. **Forward chaining** is data-driven: start from facts, fire matching rules, derive all consequences (good for monitoring). **Backward chaining** is goal-driven: start from a goal, find rules that conclude it, and prove the subgoals. **Backward chaining is more efficient when there is a specific query/goal** (e.g. "Does this patient have malaria?"), avoiding deriving many irrelevant facts.

Q24. Deductive vs inductive vs abductive reasoning + use case.

Type	Direction	AI use case
Deductive	General → specific (certain if premises true)	Rule-based expert systems, logic
Inductive	Specific → general (probable)	Machine-learning generalization from examples
Abductive	Observation → best explanation	Diagnostic/medical systems

Q25. Semantic networks & frames; inheritance/instances/defaults; limits.

Semantic networks use nodes (concepts) and labelled edges (is-a, has-a); **inheritance** flows along is-a links. **Frames** are structured records of **slots** (attributes) with **default values**; instance frames inherit from class frames and slot defaults fill unknowns. **Limitations vs formal logic:** no precise semantics, and difficulty expressing quantification, negation and disjunction, plus ambiguous multiple inheritance.

Module 5 — Machine Learning (Q26-Q33)

Q26. ML; vs deterministic programming; why suited to perception.

ML learns patterns from data to make predictions without being explicitly programmed with rules. Traditional programming: human writes rules (input + program → output); **ML inverts this** (input + output → learns the model). It suits perceptual/high-dimensional tasks (vision, speech) because hand-coding the rules is infeasible, yet abundant data exists to learn from.

Q27. Three ML paradigms + real-world app + justification.

Paradigm	Application & why
Supervised	Medical image diagnosis — labelled scans, needs an accurate input→label mapping
Unsupervised	Customer segmentation — no labels; discover hidden groupings
Reinforcement	Game-playing/robotics (AlphaGo) — sequential decisions learned from rewards, no "correct" label per move

Q28. Bias-variance trade-off.

Expected error ≈ **bias² + variance + irreducible noise**. **High bias** = model too simple → **underfitting**, high error on both training and test data (systematic error). **High variance** = model too complex → **overfitting**, low training error but high test error (over-sensitive to the data). Increasing complexity lowers bias but raises variance; the goal is to balance the two.

Q29. Overfitting vs underfitting + prevention.

Overfitting: model memorizes noise — great on training, poor on new data. **Underfitting**: too simple — poor on both. **Prevent overfitting**: cross-validation, regularization (L1/L2), pruning (trees), dropout (neural nets), early stopping, more/clean data, simpler model.

Q30. Classification vs regression + metrics.

Classification predicts a discrete category (spam/not-spam) — metrics: **accuracy, precision, recall, F1-score**, confusion matrix. **Regression** predicts a continuous value (house price) — metrics: **MSE, RMSE, MAE, R²**.

Q31. KNN; effect of K and distance metric.

KNN classifies a point by the majority vote of its **K nearest neighbours** (lazy, instance-based). **Small K** → noisy, jagged boundary (overfits); **large K** → smoother boundary (may underfit). The **distance metric** (Euclidean vs Manhattan) defines "nearest" and reshapes the boundary; features must be scaled for fair distances.

Q32. K-means; elbow method; centroid sensitivity.

K-means: pick K, initialise centroids, assign each point to the nearest centroid, recompute centroids, repeat until convergence. **Elbow method**: plot within-cluster sum of squares against K and choose K at the "elbow" where the curve flattens. It is **sensitive to initial centroids** (can hit a local optimum) — mitigated by **k-means++** initialisation or multiple restarts.

Q33. Reinforcement Learning; components; cumulative vs immediate reward.

RL: an agent learns optimal behaviour by trial-and-error interaction. Components: **Agent, Environment, State, Action, Reward, Policy** (state → action). The goal is to maximize **expected cumulative (long-term) reward**, not immediate reward — so the agent may sacrifice short-term gain for a larger long-term return (discount factor γ weights future rewards).

Module 6 — Neural Networks & Deep Learning (Q34-Q38)

Q34. Artificial Neural Network; biological inspiration; neuron maths.

An **ANN** is a network of interconnected artificial neurons, inspired by biological neurons (dendrites = inputs, synapses = weights, axon = output). A single neuron computes a **weighted sum $z = \sum(w_i x_i) + b$** , then applies an **activation function**: output = f(z).

Q35. Perceptron; XOR limitation; Minsky & Papert.

A **perceptron** is a single-layer linear classifier: output = step($\sum w x + b$). It can only separate **linearly separable** data, so it **cannot learn XOR**. Minsky & Papert (1969) proved this limitation, which collapsed confidence and funding in neural networks and helped trigger an AI winter.

Q36. MLP; hidden layers; Universal Approximation.

A **Multi-Layer Perceptron** has input, one or more **hidden layers**, and output, with non-linear activations. Hidden layers let it form **non-linear decision boundaries** (solving XOR). The **Universal Approximation Theorem** states an MLP with one sufficiently large hidden layer can approximate any continuous function.

Q37. Activation functions: Sigmoid, Tanh, ReLU; ReLU & vanishing gradient.

Function	Range / use	Issue
Sigmoid	(0,1) — output probabilities	Saturates → vanishing gradients
Tanh	(−1,1) — zero-centred	Still saturates
ReLU = max(0,x)	Hidden layers of deep nets	Cheap, non-saturating → fixes vanishing gradient (but "dying ReLU")

Q38. Backpropagation; chain rule; learning rate.

Backpropagation computes the gradient of the loss with respect to every weight by propagating the error backward layer by layer using the **chain rule**, then gradient descent updates each weight: $w \leftarrow w - \eta \cdot \partial L / \partial w$. The **learning rate η** sets the step size — too high diverges/oscillates, too low converges slowly.

Module 7 — NLP & Computer Vision (Q39-Q43)

Q39. NLP; four challenges.

NLP enables computers to understand and generate human language. Challenges: **lexical ambiguity** (a word with several meanings — "bank"), **syntactic ambiguity** (a sentence with multiple parses), **coreference resolution** (what "it/they" refers to), and **sarcasm/context** (literal meaning ≠ intended meaning).

Q40. Bag-of-Words & TF-IDF; strengths and weakness.

BoW represents a document as a vector of word counts, ignoring order. **TF-IDF** weights each term by term frequency $\times$ inverse document frequency, boosting distinctive words and damping common ones. **Strengths:** simple, fast, effective for classification/retrieval.

Weakness: they **ignore word order and semantics** (synonyms, context), giving sparse high-dimensional vectors with no real meaning.

Q41. Syntax vs semantics; parsing; NER.

Syntax = grammatical structure; **semantics** = meaning. **Syntactic parsing** builds a parse tree exposing structure (subject-verb-object).

Named Entity Recognition (NER) tags entities (person, place, organization, date), extracting **semantic** information used for understanding and information extraction.

Q42. Computer Vision; intrinsic challenges.

Computer Vision lets machines interpret images/video. Intrinsic challenges: **illumination variation**, **occlusion** (partly hidden objects), **viewpoint/scale variance**, background clutter, deformation, and intra-class variation.

Q43. CNN layers.

CNNs are built for grid-like image data. **Convolutional layer:** applies learnable filters to extract local features (edges $\rightarrow$ textures $\rightarrow$ shapes), using parameter sharing while preserving spatial structure. **Pooling layer:** downsamples (max/average) $\rightarrow$ dimensionality reduction and translation invariance. **Fully-connected layer:** flattens features to produce the final **classification**.

Module 8 — Ethics, Safety & the Future (Q44-Q50)

Q44. Major ethical concerns; how they interconnect.

Algorithmic bias (unfair outcomes), **mass surveillance** (privacy erosion), **autonomous weapons** (machines making lethal decisions), and **job displacement** (automation). They are interconnected: opaque, biased models deployed at scale concentrate power and amplify harm — surveillance feeds biased systems, and automation concentrates economic control.

Q45. Algorithmic bias; example; mitigation.

Algorithmic bias is systematic, unfair discrimination arising from skewed data or model design. **Example:** a hiring model trained on a male-dominated history learns to down-rank women; or recidivism scoring (COMPAS) over-predicts risk for minorities. **Mitigation:** diverse representative data, bias audits and fairness metrics, transparency/XAI, human oversight, and de-biasing techniques.

Q46. Explainable AI (XAI); importance; accuracy vs explainability.

XAI makes model decisions interpretable to humans. It is critical in high-stakes domains (healthcare, autonomous driving) for **trust**, **accountability**, **debugging and legal compliance**. **Trade-off:** complex black-box models (deep nets) are accurate but opaque, while simpler models (decision trees, linear) are interpretable but may be less accurate; tools like LIME/SHAP explain black boxes.

Q47. Short- vs long-term future; bottlenecks to AGI.

Short-term: stronger LLMs, pervasive automation, specialised domain AI (medicine, science). **Long-term:** gradual progress toward broader capability. **Bottlenecks:** huge data requirements, energy/compute cost, **lack of common-sense reasoning**, brittleness, and the unsolved alignment/safety problem.

Q48. The singularity (Kurzweil); critical analysis.

The **technological singularity** is a hypothetical point at which AI surpasses human intelligence and recursively self-improves, causing explosive, uncontrollable growth; Kurzweil predicts ~ 2045 via exponential technology trends. **Critique:** Moore's law is slowing, we still don't understand general intelligence or consciousness, today's AI is narrow, and exponential curves often flatten into S-curves — so a within-century singularity remains speculative.

Q49. AI transforming an industry + risks (personalized medicine).

In **personalized medicine**, AI analyses genomics and patient records to tailor treatments, predict disease risk, and accelerate drug discovery. **Risks/limitations:** data privacy, biased/unrepresentative datasets, regulatory and liability issues, over-reliance on opaque models, and the need for rigorous clinical validation.

Q50. "Robust" and "safe" AI; multi-layered framework.

Robust = reliable under novel, noisy or adversarial inputs; **safe** = avoids harmful actions and fails gracefully. A multi-layered framework: **Technical** (rigorous testing, adversarial robustness, uncertainty estimation, fail-safes, interpretability), **Regulatory** (standards, audits, certification, clear accountability), and **Ethical** (human-in-the-loop oversight, value alignment, transparency, fairness). For critical national infrastructure add redundancy and continuous monitoring before deployment.

Short Questions (Q51-Q57)

Q51. Three applications of AI.

Healthcare — disease diagnosis from scans; **Autonomous vehicles** — self-driving perception & control; **Virtual assistants/chatbots** — answer queries in natural language (also fraud detection, recommendations).

Q52. ML + (a) supervised (b) unsupervised.

ML = learning patterns from data to predict/decide without explicit programming. **(a) Supervised** — learns from **labelled** data an input $\rightarrow$ output mapping (e.g. spam classification). **(b) Unsupervised** — finds structure in **unlabelled** data (e.g. clustering customers).

Q53. Expert system + two components.

A program that emulates a human expert's decision-making. Two major components: **Knowledge Base** (facts & rules) and **Inference Engine** (applies rules to reason). [Often also a user interface.]

Q54. Narrow vs General AI + examples.

Narrow AI handles one task (Siri, spam filter, chess engine). **General AI** would match human ability across any task (hypothetical, not yet achieved).

Q55. NLP + three applications.

NLP = machines processing human language. Apps: **machine translation**, **sentiment analysis**, **chatbots/speech recognition**.

Q56. Three advantages and three disadvantages of AI.

Advantages: automation/efficiency; works 24/7 without fatigue; reduces human error / handles big data. **Disadvantages:** high cost; job displacement; lacks creativity & common sense (plus bias and privacy risks).

Q57. AI in healthcare, education, banking.

Healthcare: diagnosis & drug discovery. **Education:** personalized tutoring & automated grading. **Banking:** fraud detection & credit scoring.

Specific Problems (1-6)

SP1. Intelligent agent + two characteristics.

An entity that perceives its environment (sensors) and acts on it (actuators). Characteristics: **autonomous** and **goal-directed/rational** (also reactive, able to perceive & act).

SP2. Supervised vs Reinforcement learning.

Supervised learns from **labelled examples** (correct answers given). **Reinforcement** learns by **trial and error from rewards/penalties**, with no labelled answers.

SP3. Expert system + two areas of use.

A system emulating expert decision-making. Used in **medical diagnosis** (e.g. MYCIN) and **financial/credit analysis** (also fault diagnosis).

SP4. NLP + two applications.

NLP = processing human language. Applications: **machine translation** and **chatbots/virtual assistants**.

SP5. Three ethical issues of AI.

Algorithmic bias, privacy/surveillance, job displacement (also accountability and autonomous weapons).

SP6. Importance of AI in (a) Agriculture (b) Security (c) Transportation.

Agriculture: crop monitoring, yield/rainfall prediction, disease detection. **Security:** surveillance, intrusion/fraud detection, facial recognition. **Transportation:** self-driving vehicles, traffic optimization, route planning.

Problem Scenarios

Scenario 1 — AI in Agriculture (10)

a) AI improves productivity by monitoring crop health, predicting rainfall/weather, detecting plant diseases early and optimizing irrigation/fertilizer → higher, healthier yields. **b)** Two technologies: **computer vision** (drone imagery for disease detection) and **machine-learning prediction** (rainfall/yield) — with IoT sensors. **c)** Benefits: early disease detection, efficient resource use, higher yield (and less manual labour). **d)** Challenges: high cost and lack of technical skills / poor rural connectivity.

Scenario 2 — AI-Based E-Learning (10)

a) Personalized learning = AI tailoring content and pace to each student's performance, needs and learning speed. **b)** ML analyses each student's performance data, identifies weak areas and learning speed, and recommends suitable materials (content-/collaborative-filtering). **c)** Advantages: personalization, instant feedback, 24/7 scalable access. **d)** Limitations: lacks human/emotional support; data-privacy and infrastructure requirements.

Scenario 3 — AI in Healthcare (10)

a) Suitable system: a **machine-learning diagnostic (classification) system / medical expert system**. **b)** ML improves accuracy by learning patterns from large labelled datasets of records and lab results, detecting subtle correlations humans miss and reducing error. **c)** Benefits: faster diagnosis and fewer medical errors. **d)** Challenges: data privacy/security; biased or incomplete data; cost, integration and regulatory/trust issues.

Scenario 4 — AI Chatbot for a University (10)

a) A chatbot is software that simulates human conversation via text or voice. **b)** Technology used: **Natural Language Processing** (with ML). **c)** NLP lets the chatbot understand a student's query (interpret intent from natural language) and generate an appropriate response. **d)** Advantages: 24/7 instant answers, reduced staff workload, consistent responses at scale.

Scenario 5 — AI Smart Traffic System (10)

a) AI reduces congestion by analysing real-time camera/sensor data and dynamically adjusting traffic-light timing, predicting flow and rerouting vehicles. **b)** Two technologies: **computer vision** (vehicle detection) and **ML/optimization** for signal control (with IoT sensors). **c)** Benefits: less congestion and waiting, lower emissions, improved safety and emergency response. **d)** Risks: high cost/infrastructure and system failure/cyber-attack or privacy concerns.